

MAR 6930 · Marketing Data Visualization

Designing visualization

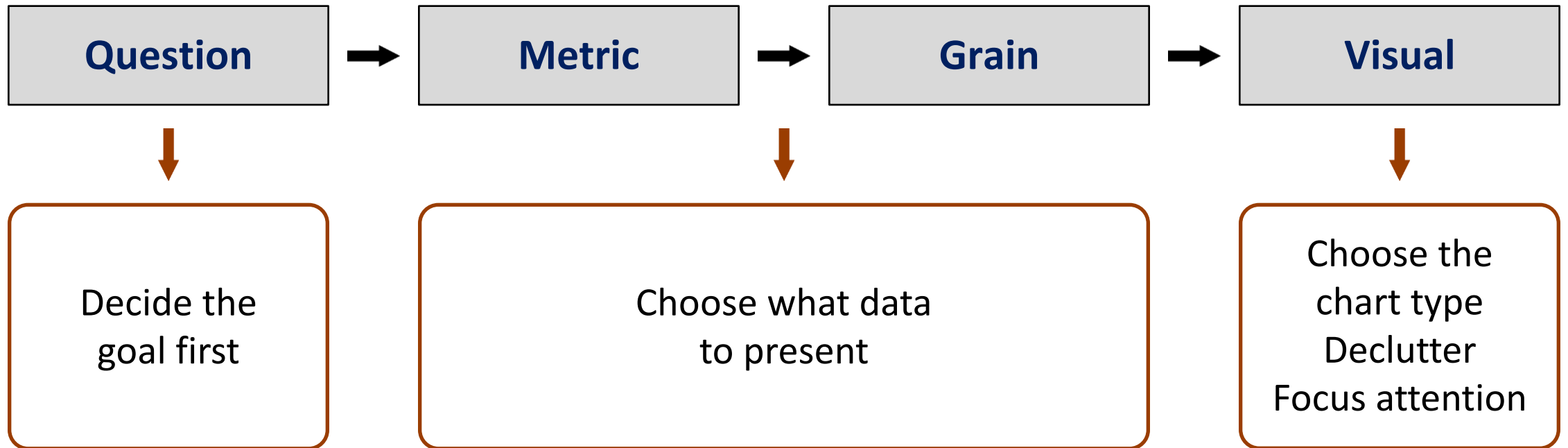
Dr. Tianxin Zou

Some materials are adapted from Knaflic (2015), *Storytelling with Data*.

Goals

1. Decide the visualization goal before making charts.
2. Choose the data and the chart type from the business question.
3. Remove clutter that does not carry the message.
4. Direct the audience's attention with color, size and enclosure.

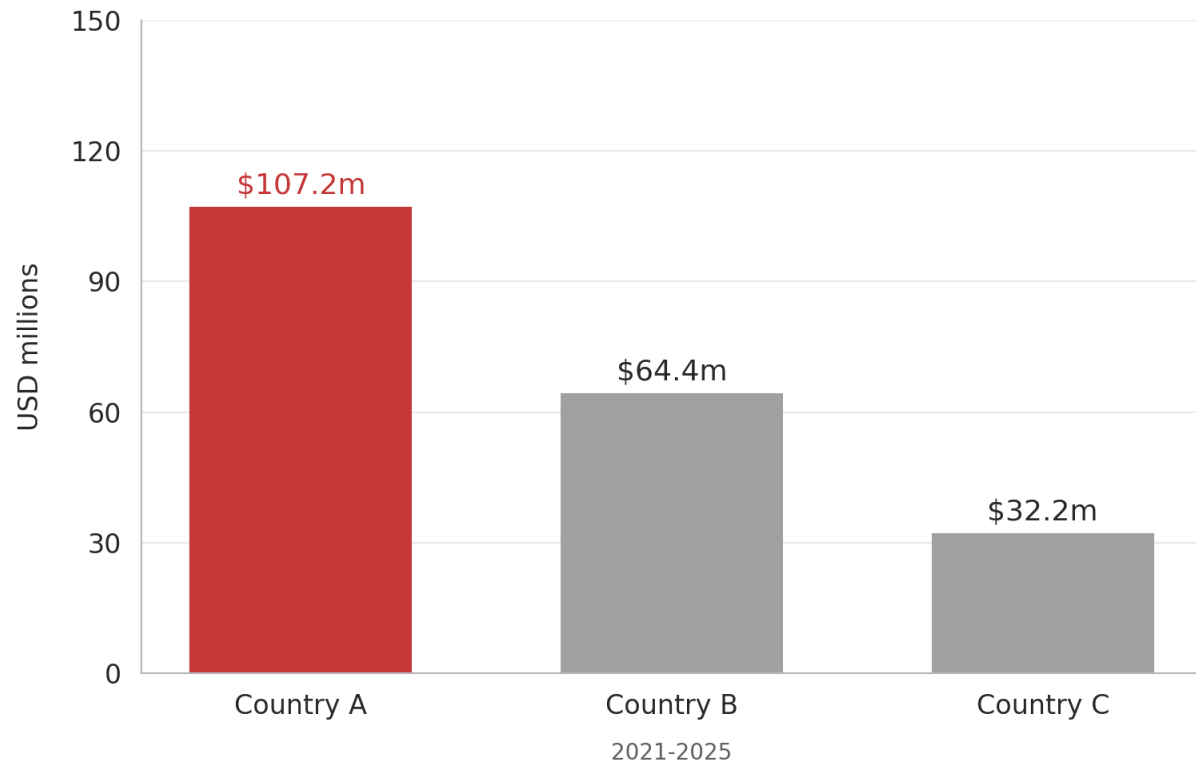
The four-step pipeline, revisited



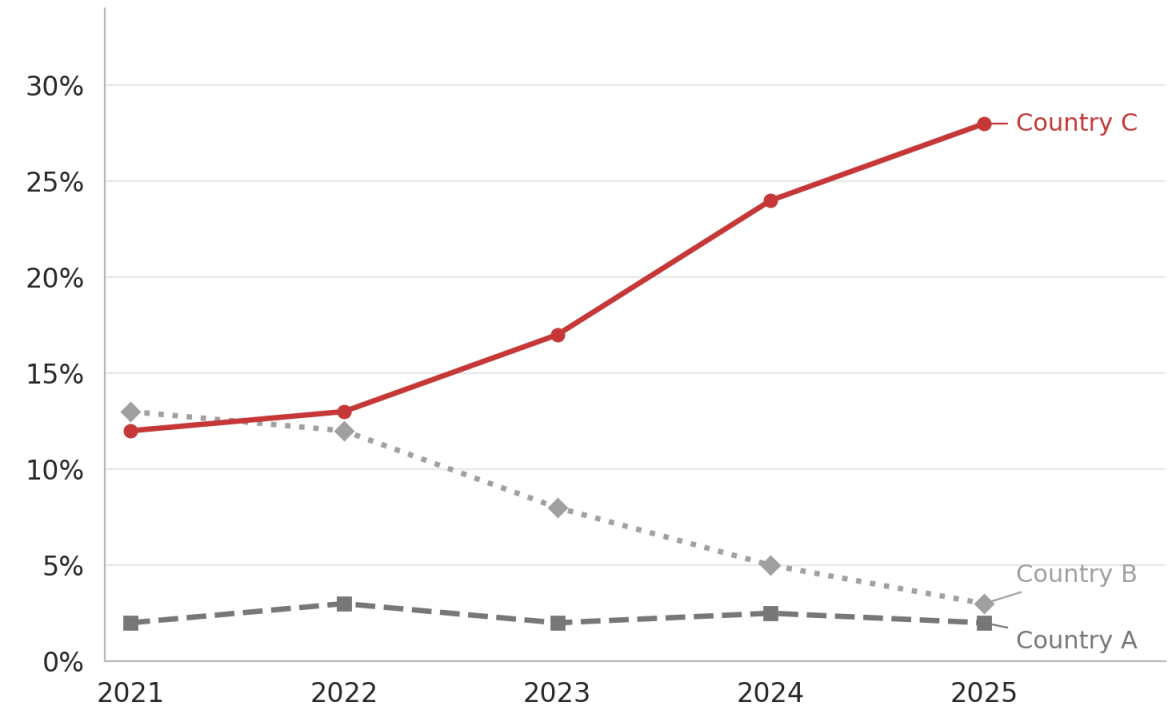
Today: general visual design principles for each of the four steps.

Which graph is better?

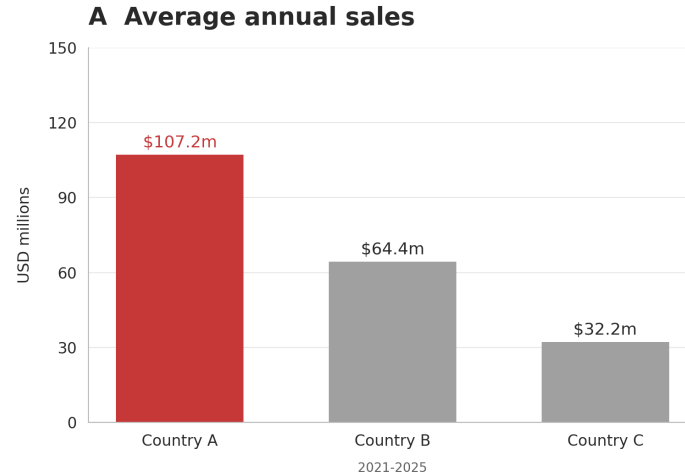
A Average annual sales



B Annual growth rate



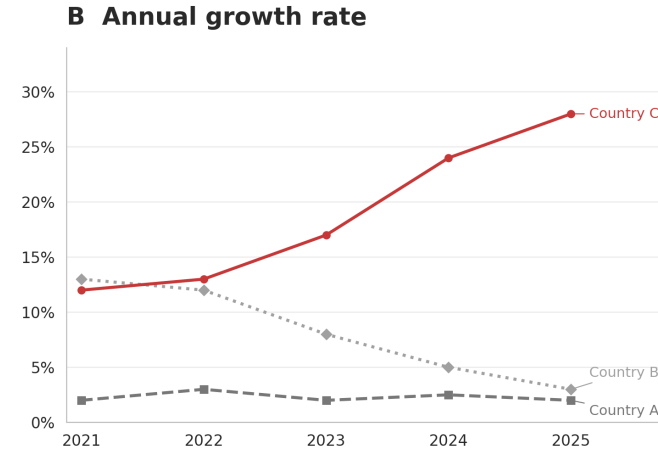
It depends on your goal



Goal 1: “Identify the market contributing the most annual sales.”

Graph A fits

A is the largest market, B about 60% and C about 30% of A.



Goal 2: “Identify a market where growth is accelerating.”

Graph B fits

C's growth rises from 12% to 28% while B's slows from 13% to 3% and A stays at 2–3%.

Takeaway: which graph is better depends on the question you need answered.

Before you draw: whom, what, how

Whom

Who is the audience or decision maker?

What

What do you need them to know or do?

How

What data makes the case?

When you ask your team to build a communication, answer up front:

- What background is relevant or essential?
- Who is the audience or decision maker, and what do we know about them?
- What biases might make them supportive of or resistant to the message?
- What data is available to strengthen the case, and is it familiar or new to them?
- Where are the risks that could weaken the case, and should we address them proactively?

Example: a summer science program

Situation: you are a fourth-grade science teacher. You ran a pilot summer program on science and surveyed the children at the start and at the end. The data looks like a success, and you want to keep offering the program.

Whom

The budget committee

Controls the funding needed to continue the program.

Future student participants

Decide whether they want to join next summer.

What

The program was a success; please **approve a budget of \$X** to continue it.

You will **discover the fun of science and make new friends**; join us next summer.

How

Science scores on standardized exams: participants against non-participants, before and after the program.

Survey results from last summer's students: the share who said science was fun and that they made new friends.

Takeaway: same program; the audience changes the ask and the data you show.

Exercise: which data answers the question?

1. Which state has the fastest sales growth?

A. City-level daily sales,
last three years

Too fine a grain: cities and days bury the state trend.

B. State-level quarterly sales,
last three years

C. State-level total sales,
last quarter only

One period only: it shows size, not growth.

2. Which product category contributes most to our revenue?

A. Revenue by product category,
this period

B. Every transaction,
by SKU

Too fine a grain: thousands of rows hide the categories.

C. Units sold
by product category

Wrong metric: units are not revenue.

3. Which customer segment responds most to our email promotions?

A. Number of responses
by segment

Wrong metric: large segments get more responses; use a rate.

B. One company-wide
response rate

Too coarse a grain: no segments to compare.

C. Response rate
by segment

Three questions to ask of every chart

1. What business question does it answer?

2. What does it make easy to see?

- What data and chart type to use?
- How to design the visual?

3. What does it hide?

- Understand the limitations of your chart: what can and what cannot be supported?
- Highlighting some information de-emphasizes the rest. Does that mislead?

Common chart types

Simply showing the figure: does “91% of customers liked the product” need a chart?

91%

of customers liked the product

- When one number carries the key message, show it big with a few words.
- Supply the context needed to interpret it: who was surveyed, when, the sample size and a comparison if one matters.

“91%, up from 84% last year”

A rise of seven percentage points: are we improving?

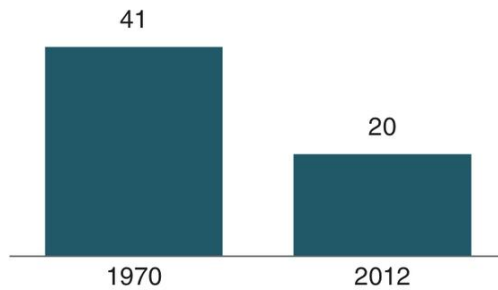
“91%, below our 95% target”

A different question: are we meeting the target?

A simple figure strips unnecessary detail

Children with a "Traditional" Stay-at-Home Mother

% of children with a married stay-at-home mother with a working husband



Note: Based on children younger than 18. Their mothers are categorized based on employment status in 1970 and 2012.

Source: Pew Research Center analysis of March Current Population Surveys Integrated Public Use Microdata Series (IPUMS-CPS), 1971 and 2013

Adapted from PEW RESEARCH CENTER

20%

of children had a
traditional stay-at-home mom
in 2012, compared to 41% in 1970

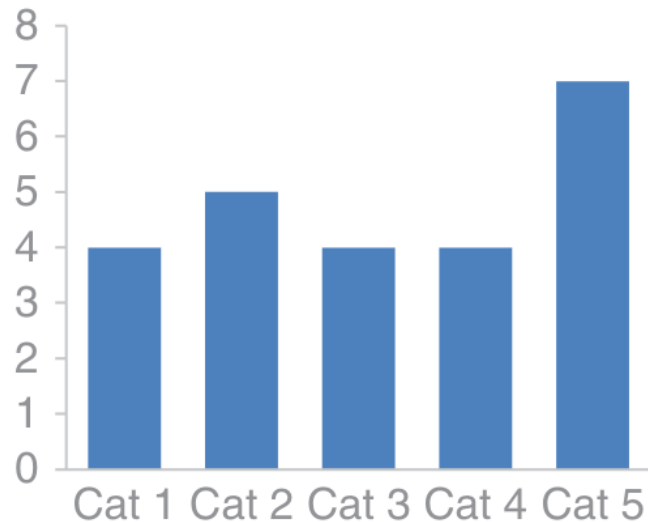
Same data, four displays, four questions

One dataset: store, month, sales, advertising spending.

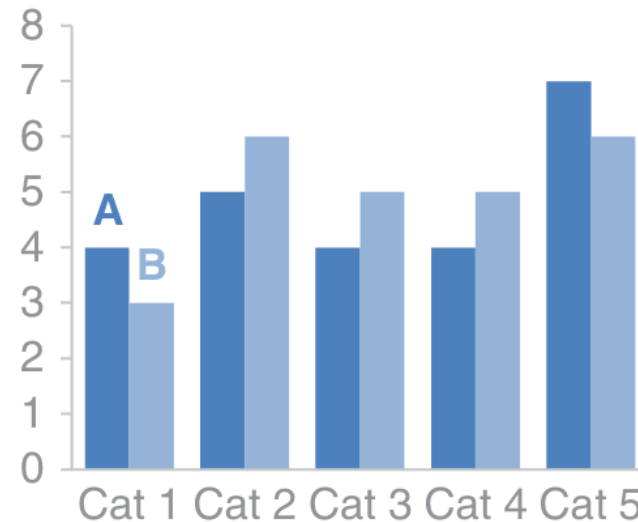
Display	Business question	Main point
Table	What were Store A's exact sales and advertising spending in March?	Gives exact values across several measures
Bar chart	Which stores had the highest March sales?	Compares magnitudes across stores
Line chart	How did Store A's sales change over the year?	Shows movement, seasonality and turning points
Scatterplot	Across stores in March, how does advertising spending relate to sales?	Shows association, clusters and unusual stores (outliers)

Bar chart: compare magnitudes across categories

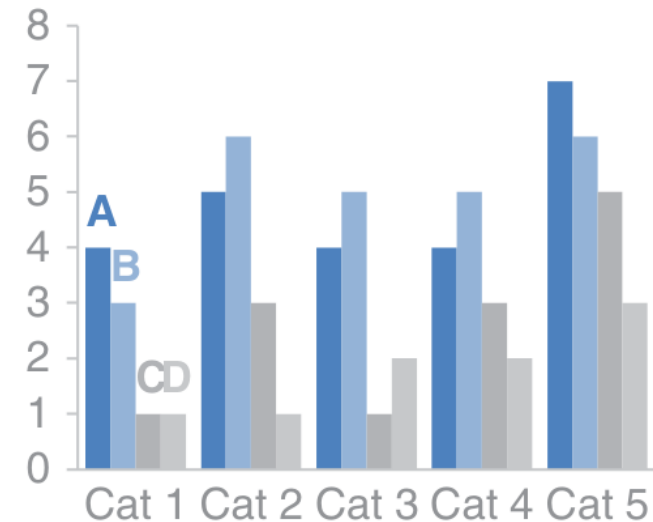
Single series



Two series



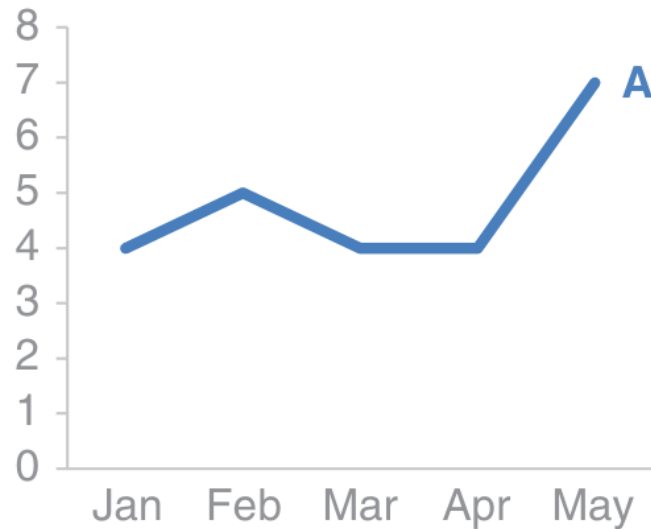
Multiple series



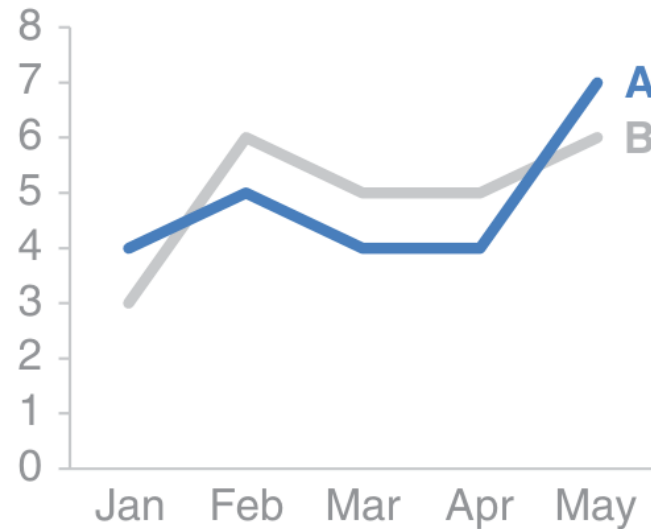
- Length on a common baseline, so the ranking is read at a glance.
- A second series still reads well; beyond a few series the comparison gets crowded.

Line chart: show movement over time

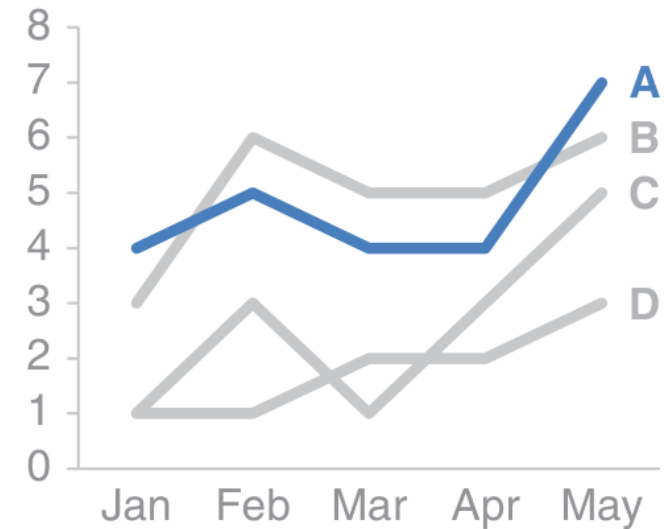
Single series



Two series



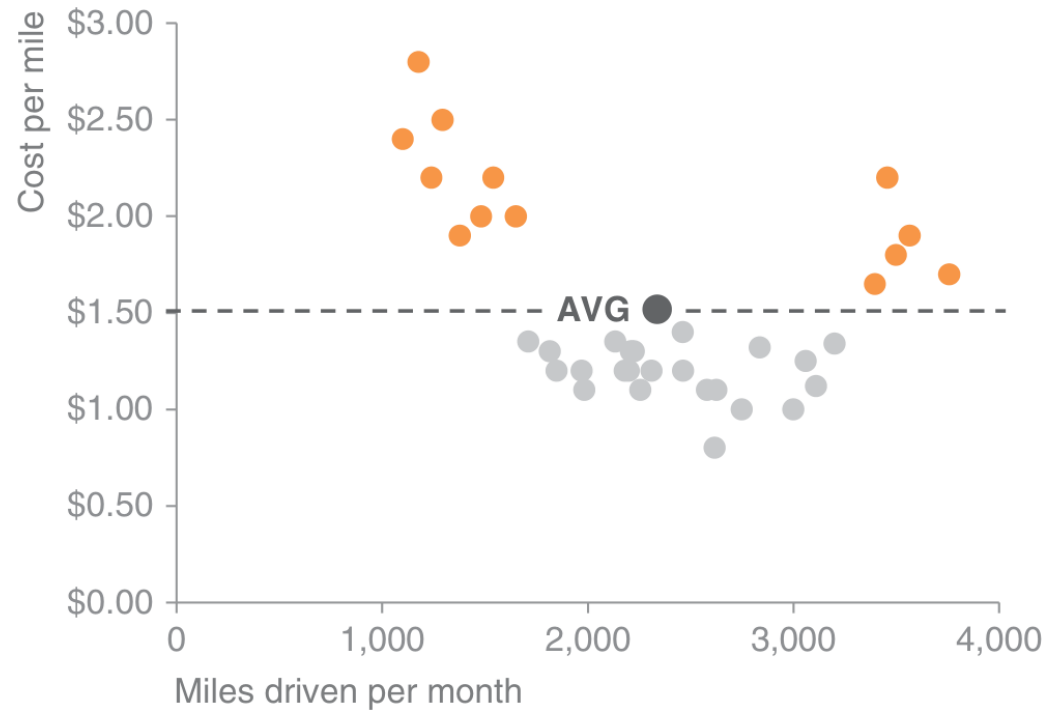
Multiple series



- Connected points imply continuity, so lines are for time.
- With multiple series, keep those you want to highlight in color and the rest gray.

Scatterplot: show how two measures relate

Cost per mile by miles driven



- Each point is one unit, here one vehicle or driver.
- The message: costs fall as miles rise, then rise again.
- To show this pattern, we add the average line and the orange color.

Slopegraph: compare two endpoints across groups

Employee feedback over time



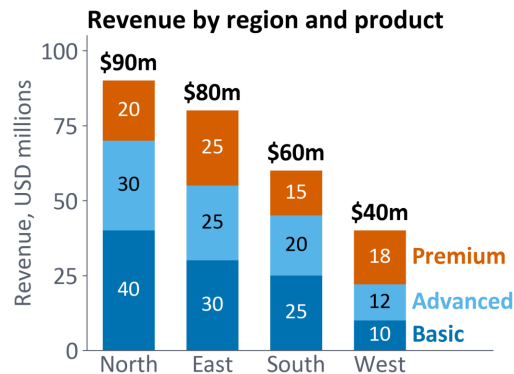
- Best for before-vs-after comparisons across several groups.
- Makes it easy to see the rate of change and the levels before and after.
- It abstracts away what happened in between, intentionally.
- You can color the segments to highlight your message.

Stacked bars: compare totals and see the breakdown

Purpose: compare totals across categories while showing how each total breaks down.

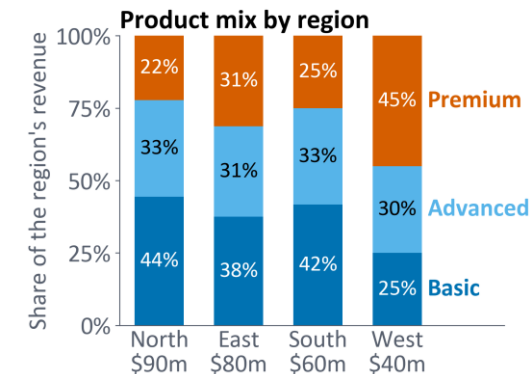
- **Example:** revenue of four regions (North, East, South, West) split into three products (Basic, Advanced, Premium).

Absolute stacks: which region earns most?



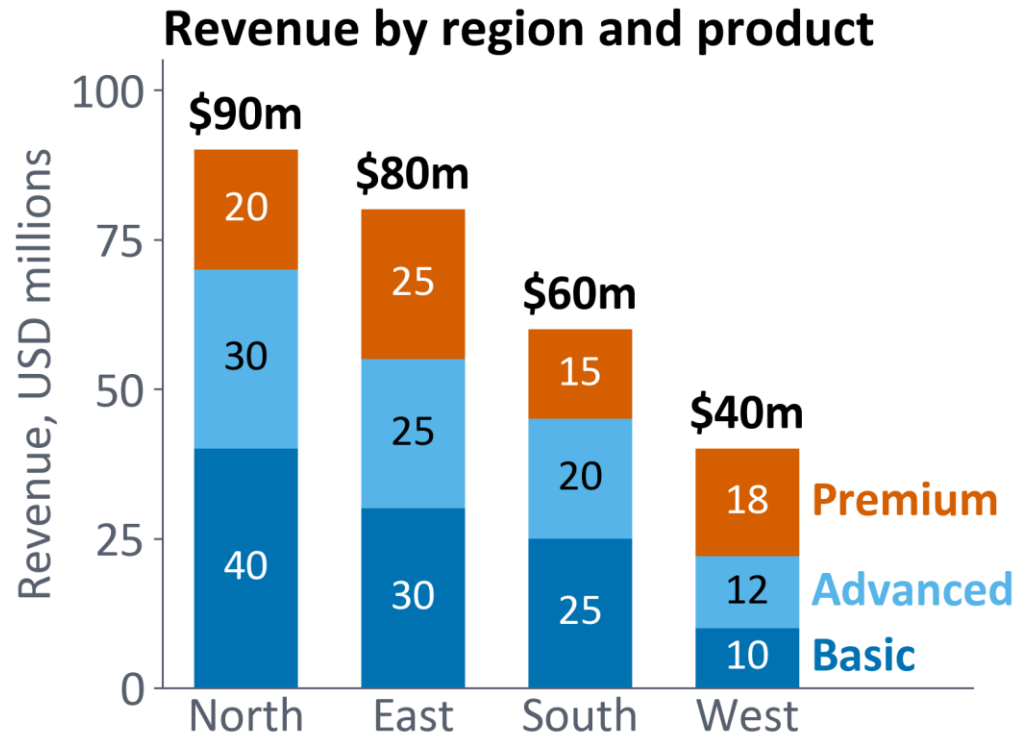
Compares **absolute values**.

Proportion stacks: which region relies most on Premium?



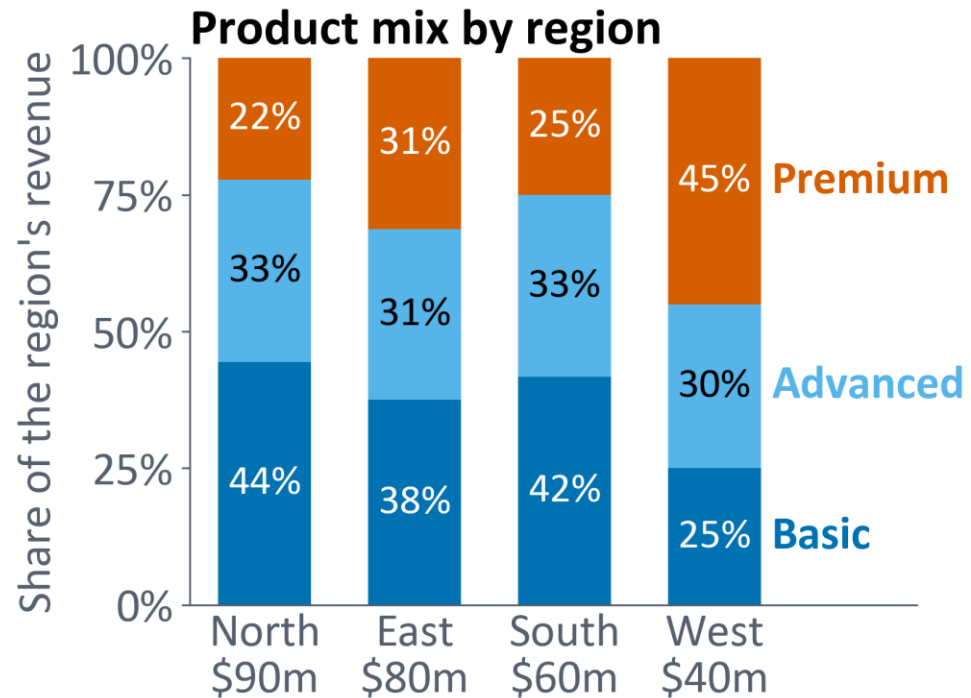
Compares **proportions**, ignoring absolute values.

Absolute stacks: which region earns most, and from what?



- Each bar's full length is the region's total.
- Segments can be compared too, with a limit:
 - Only the segment on the zero baseline (Basic) is easy to compare.
 - Advanced and Premium start at different heights.
- Stacking order:
 - Follow the categories' natural order (Basic, Advanced, Premium).
 - Or put the product that matters most on the baseline, or show it on its own.

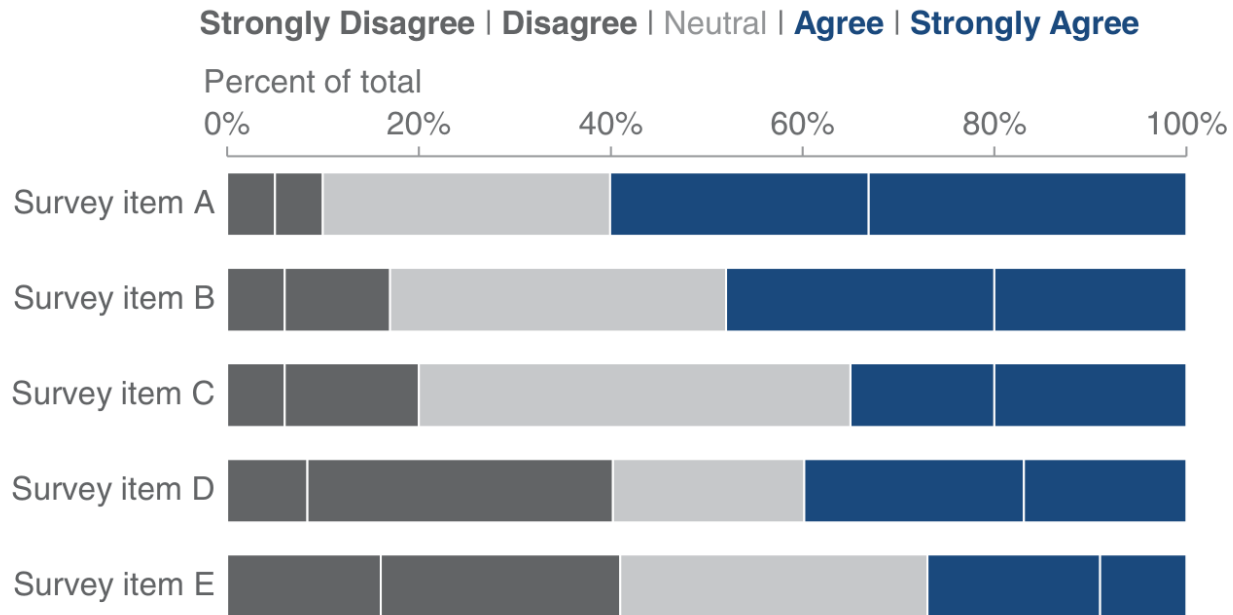
Proportion stacks: what share of each region's revenue is Premium?



- Every bar runs from 0% to 100%, so differences in proportional mix are easy to see, but differences in total disappear.
- The segments at the two ends (Basic and Premium) compare well across regions.
- The middle segment does not.

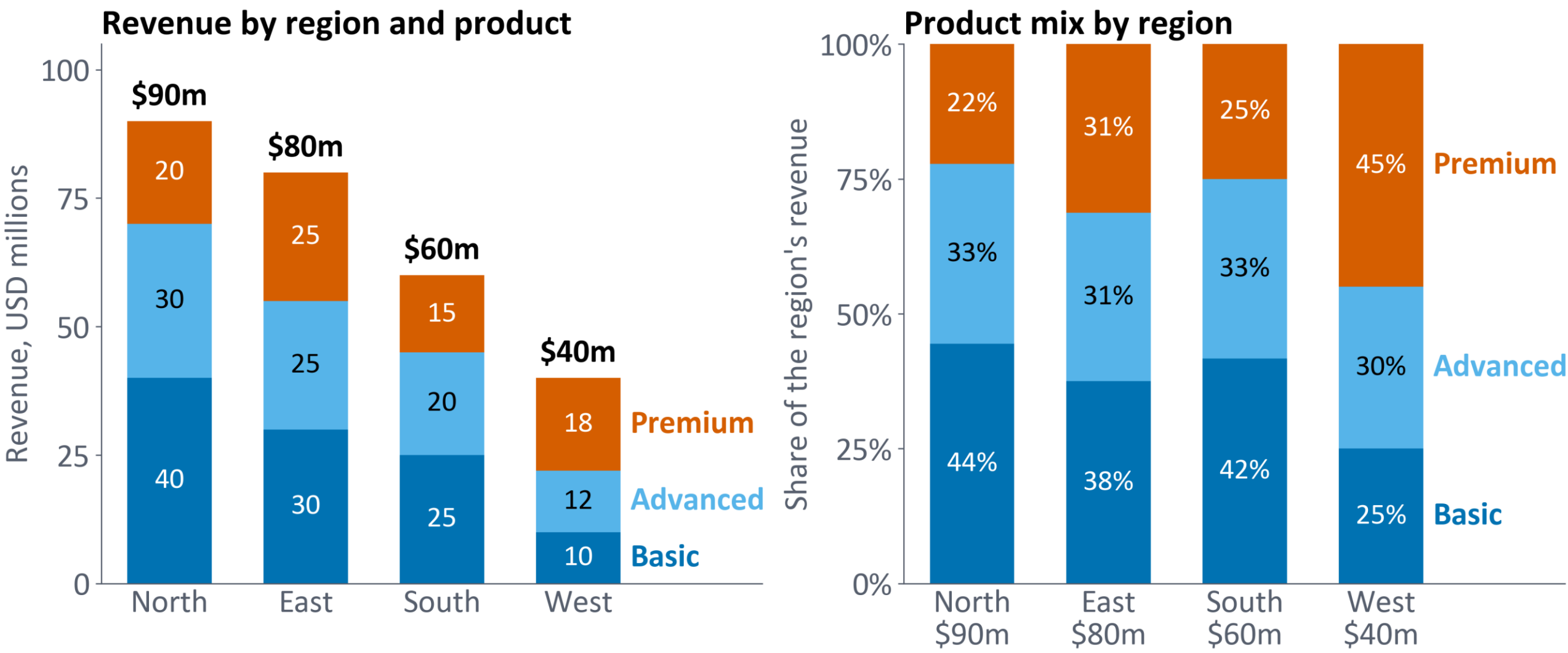
Proportion bars: often used for ordered survey answers

Survey results



- Colors can help group adjacent segments: “(strongly) disagree” and “(strongly) agree”.

Exercise: which region sells the most Premium, and which relies on it most?



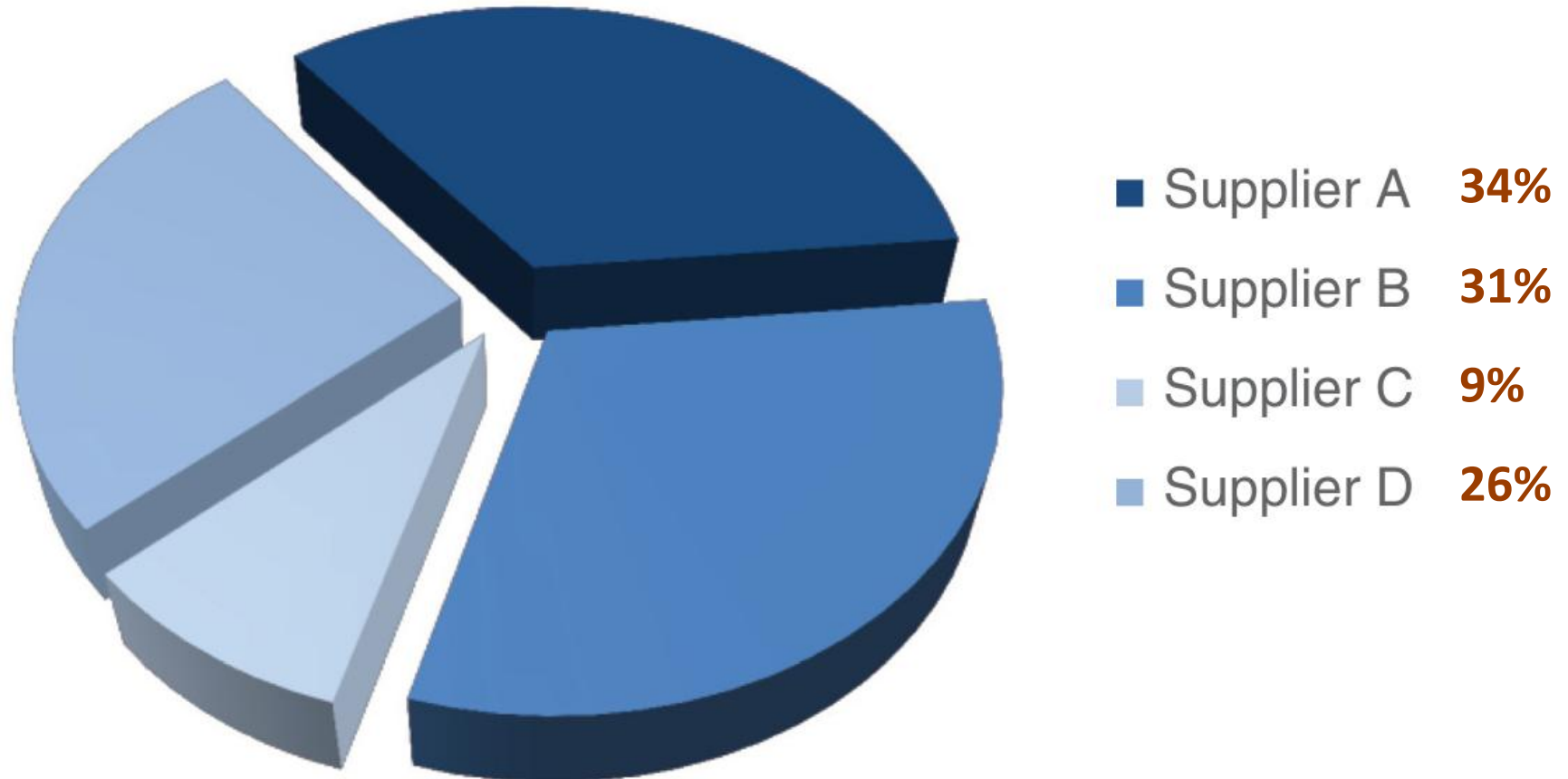
Exercise: match each business question to the best display

1. A manager needs to report one result: 91% of surveyed customers intend to renew; no breakdown or trend needed.	→ D	A. Line chart
2. A retailer wants to rank six product categories by revenue last quarter.	→ F	B. Proportion stacked bar chart
3. A retailer wants to see how monthly revenue rose and fell over two years, including seasonal peaks.	→ A	C. Scatterplot
4. A marketing analyst wants to see whether stores with higher advertising spending also had higher sales in March, and identify unusual stores.	→ C	D. Simple figure
5. A brand manager has awareness percentages for six regions before and after a campaign and wants to see which regions gained or lost ground.	→ G	E. Absolute stacked bar chart
6. A sales director wants to compare total revenue across four regions and see how much Basic, Advanced and Premium products contributed to each total.	→ E	F. Ordinary bar chart
7. A sales director wants to compare the percentage mix of Basic, Advanced and Premium across four regions, regardless of each region's total revenue.	→ B	G. Slopegraph

Chart types to avoid

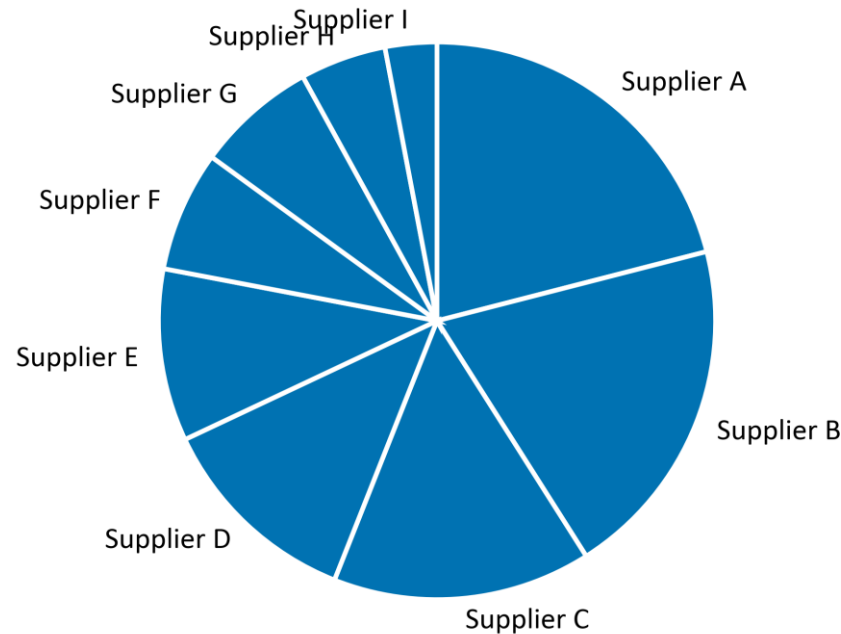
Rank the four suppliers by market share

Supplier Market Share

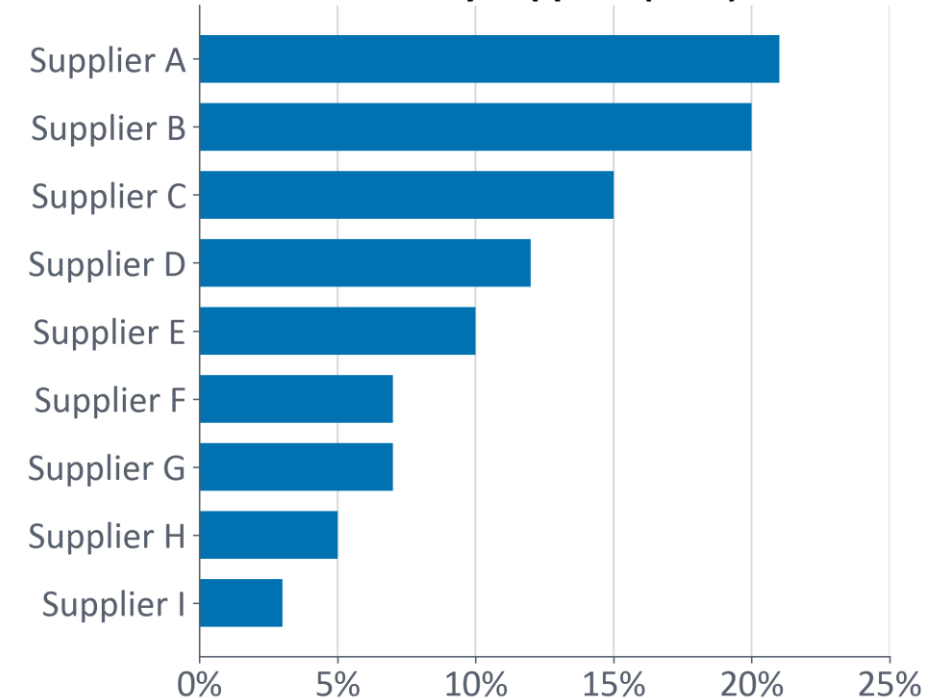


Which has a bigger market share: Supplier A or B? Supplier F or G?

Market share by supplier (pie)



Market share by supplier (bars)

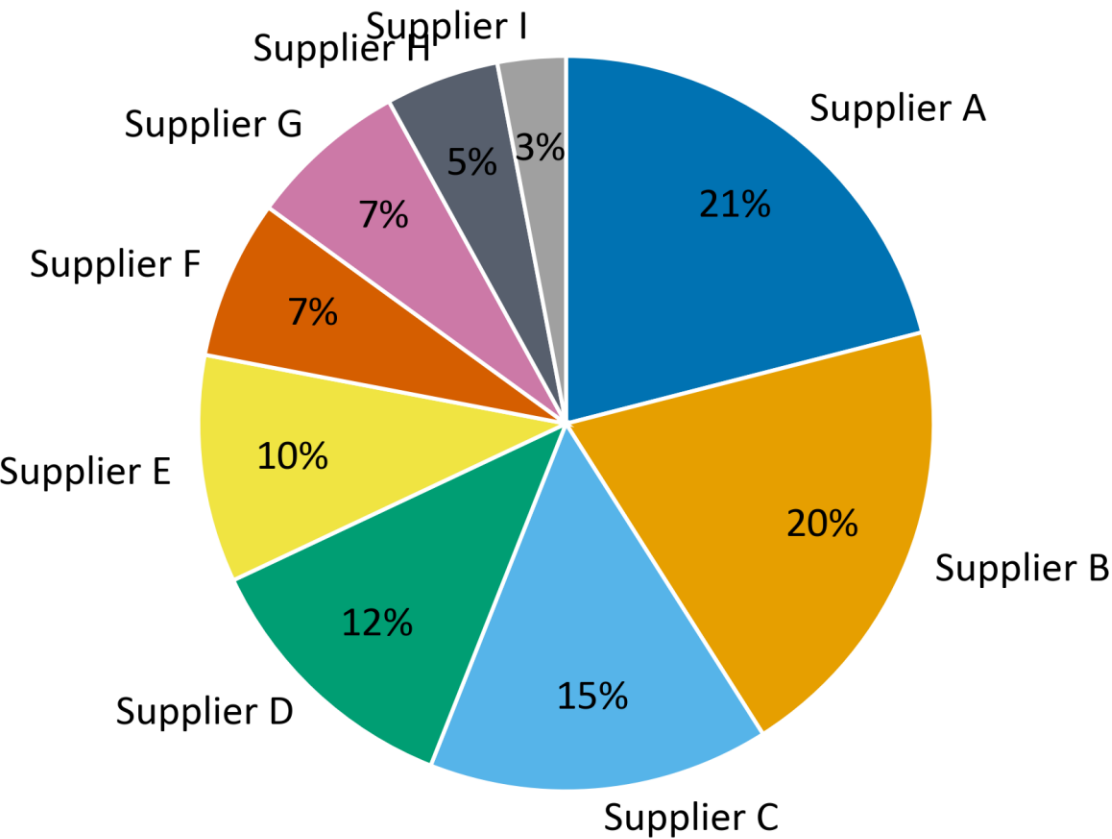


Human eyes are much better at comparing length than comparing area:

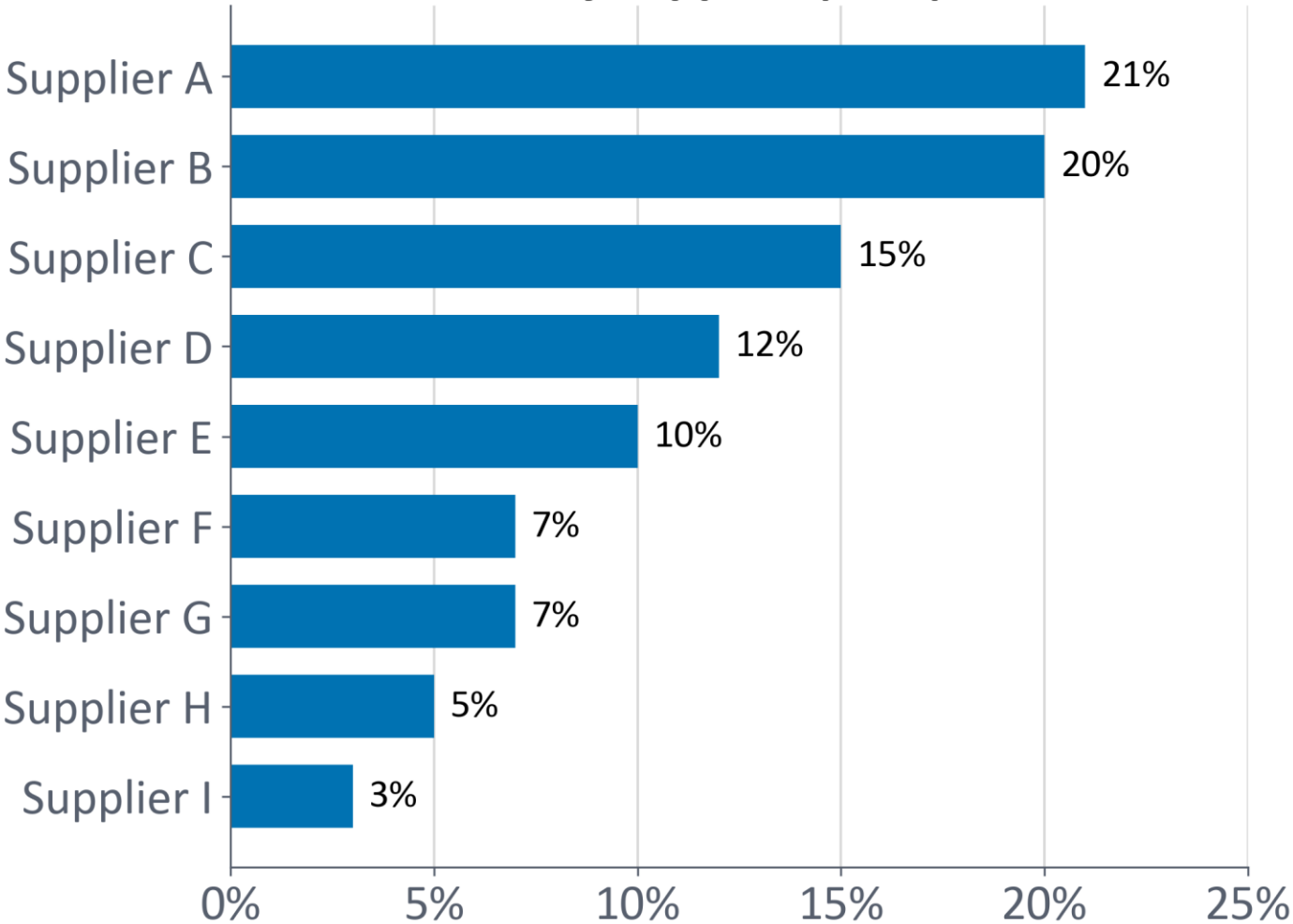
- Pie charts are hard to read: when segments are close in size, you cannot tell which is bigger.
- When they are not close, you can tell which is bigger but not by how much.

Use bars instead of pie charts: the ranking and ratios are obvious

Market share by supplier (pie)

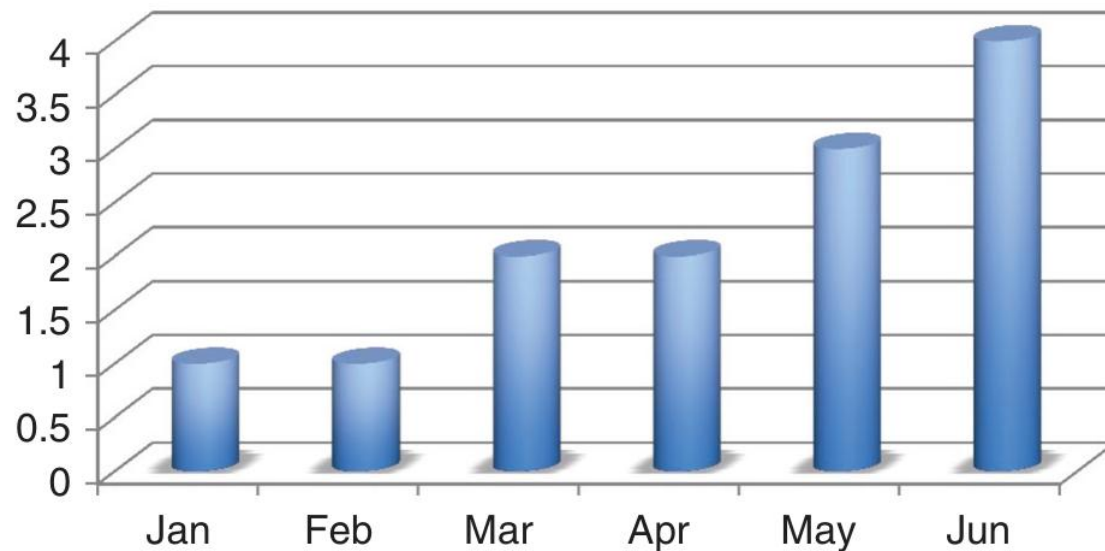


Market share by supplier (bars)



Avoid 3D visuals, especially for decorative purposes

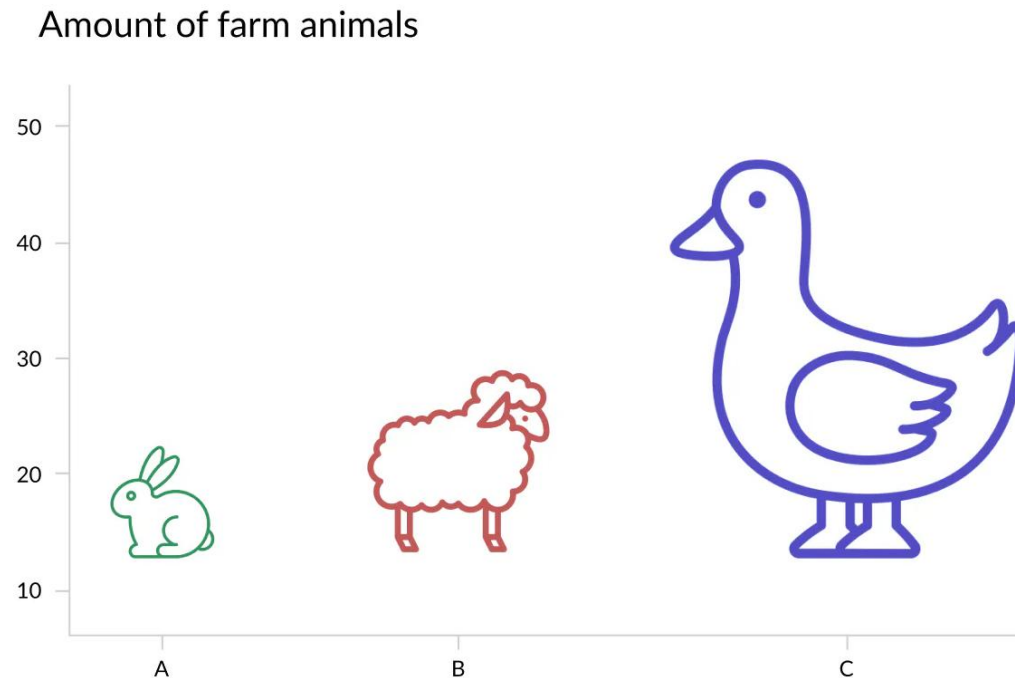
Number of issues



- January and February are 1 each; against the gridlines they read as about 0.8.
- Perspective and depth distort where bars meet the scale, and can make a nearer pie slice look larger.

Takeaway: use a 2D chart unless depth encodes a real variable (which is rare).

Remove purely decorative elements to avoid confusion



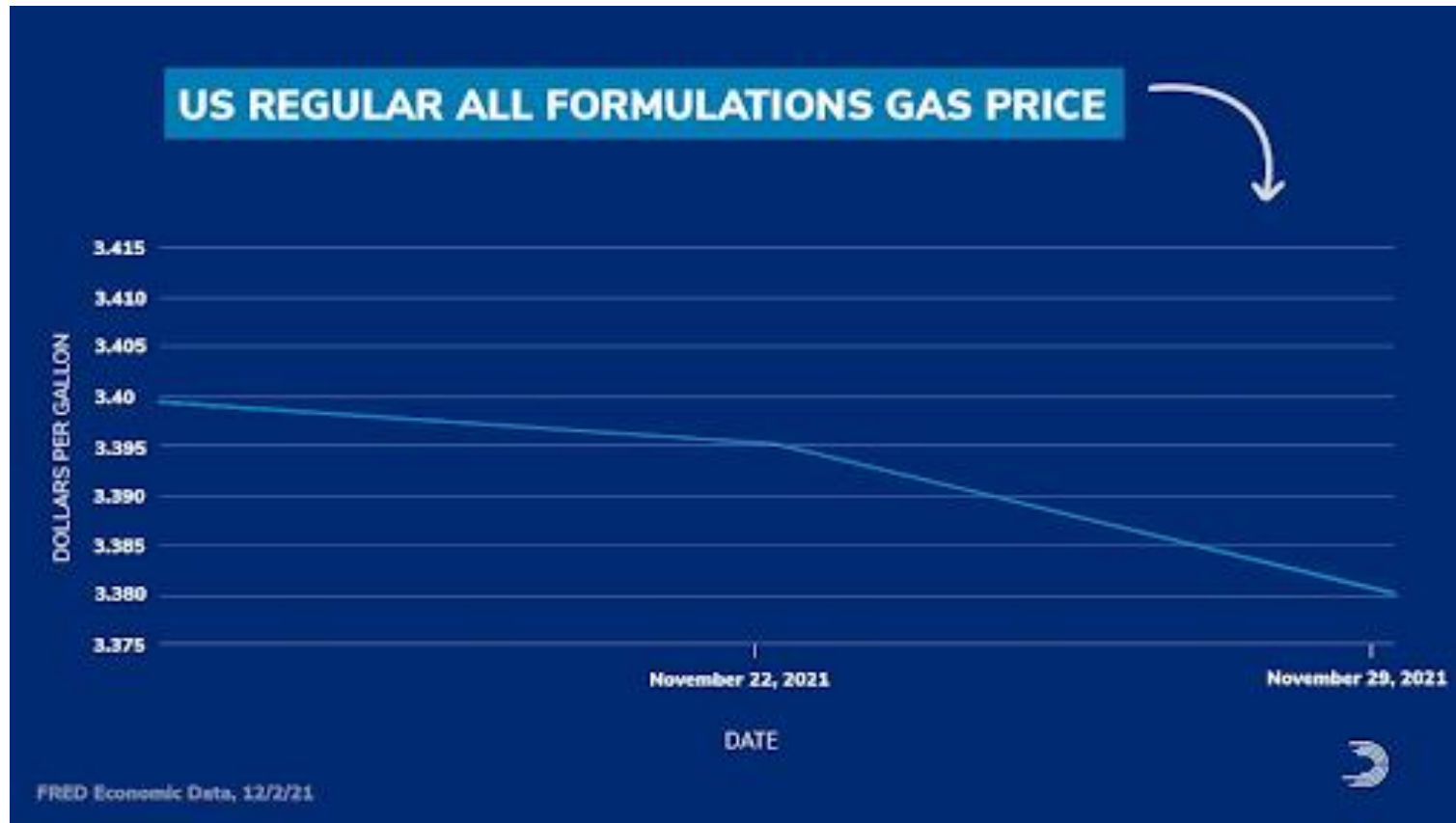
What can be confused?

- Which part shows the value: the icon's top, center, height or area?
- Different silhouettes make areas hard to compare.
- Size comparison is misleading because the icons do not start at zero.
- Doubling height and width quadruples the area, so pictures exaggerate differences.

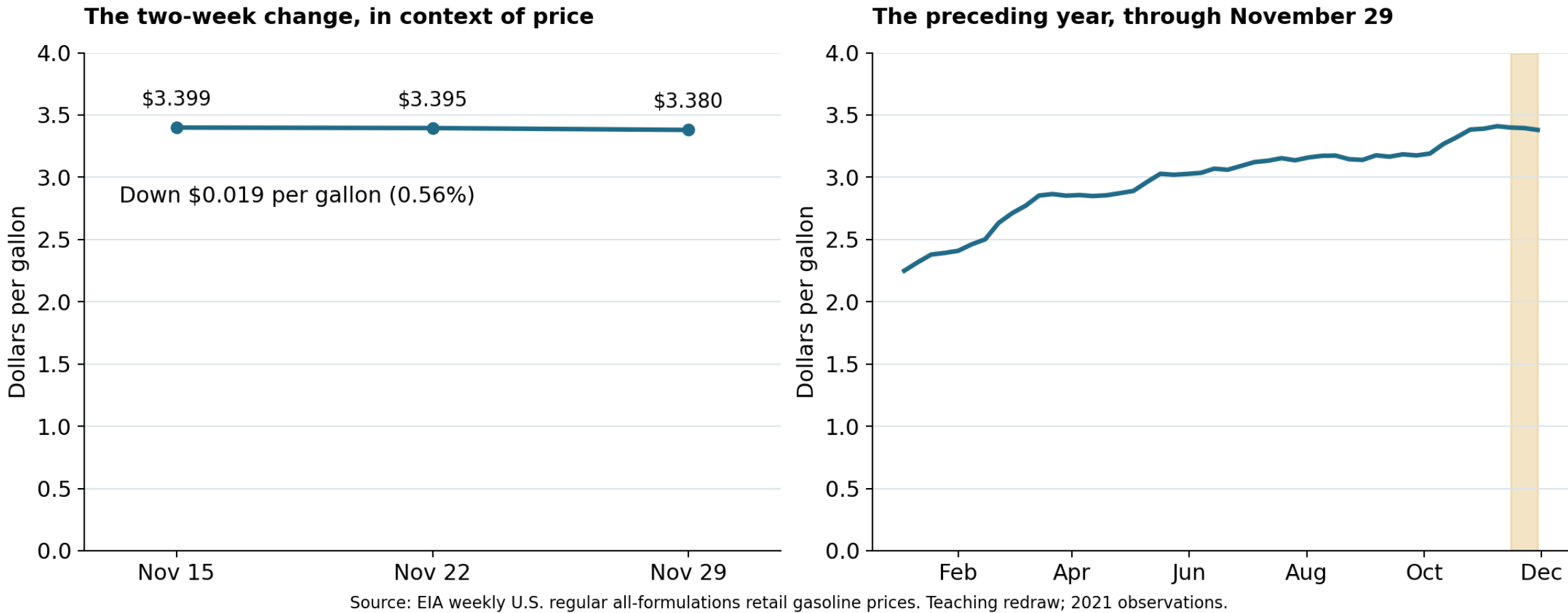
Instead: plain bars, with equal-sized icons only as labels.

Non-zero axis starts can be misleading

On December 2, 2021, the Democratic Congressional Campaign Committee (DCCC) tweeted this chart of U.S. gas prices with the message “Thanks, @JoeBiden.”



Same data, honest scale: a two-cent dip in a rising year



Build these chart types in Python

Notebook: lecture2-chart-types.ipynb with lecture2.csv

For each chart type, two plots of the same data:

1. Raw plot: explore the data

Plot the data as it comes, and look for the ranking, the trend or the pattern.

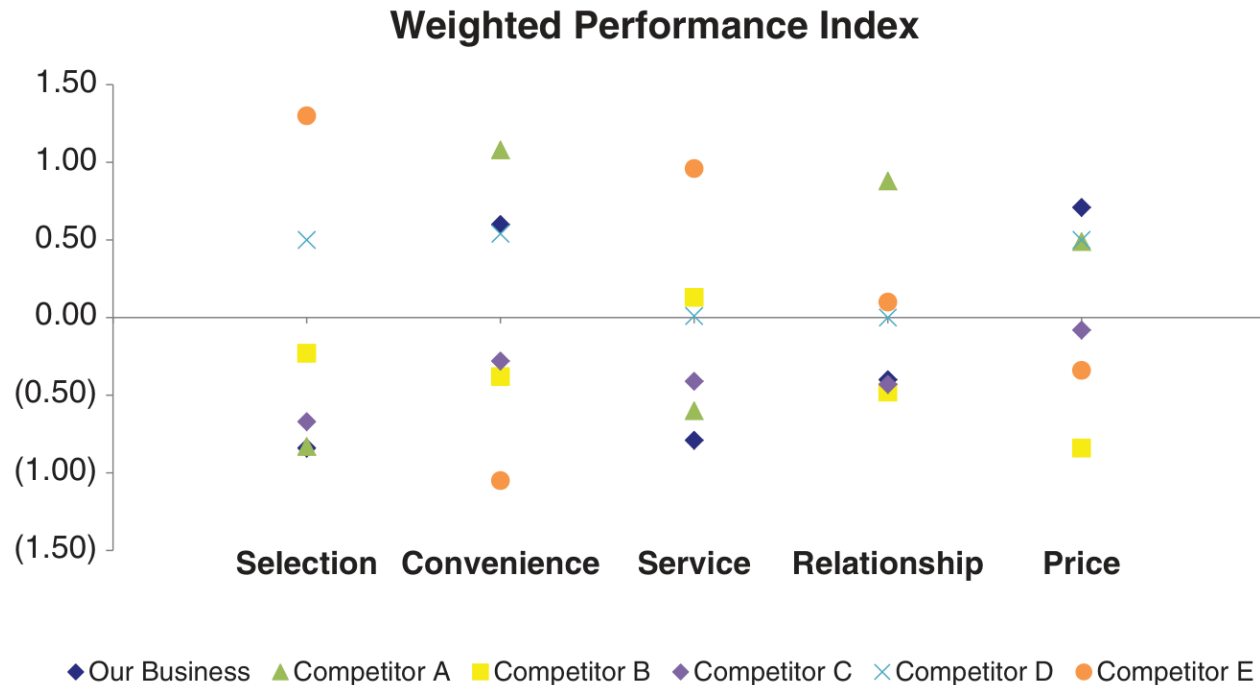
2. Polished plot: highlight the answer

Highlight the key data in color and the rest in gray, label it directly and write a title that states the answer.

Remove visual clutter

Discussion: what does this chart tell you?

Visual clutter: unnecessary elements that obscure the main message of the visual.



Too many colors and shapes

Six colors and six marker shapes for six series.

Text everywhere

A legend far from the data, bold category labels, decimals on the axis.

Negative values

The scale runs from (1.50) to 1.50, so half the points hang below zero.

No hierarchy

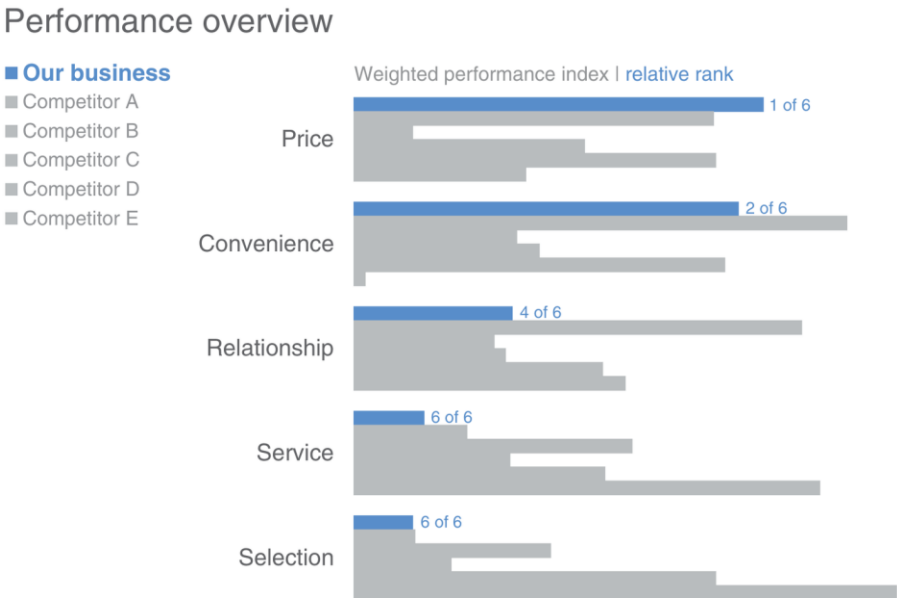
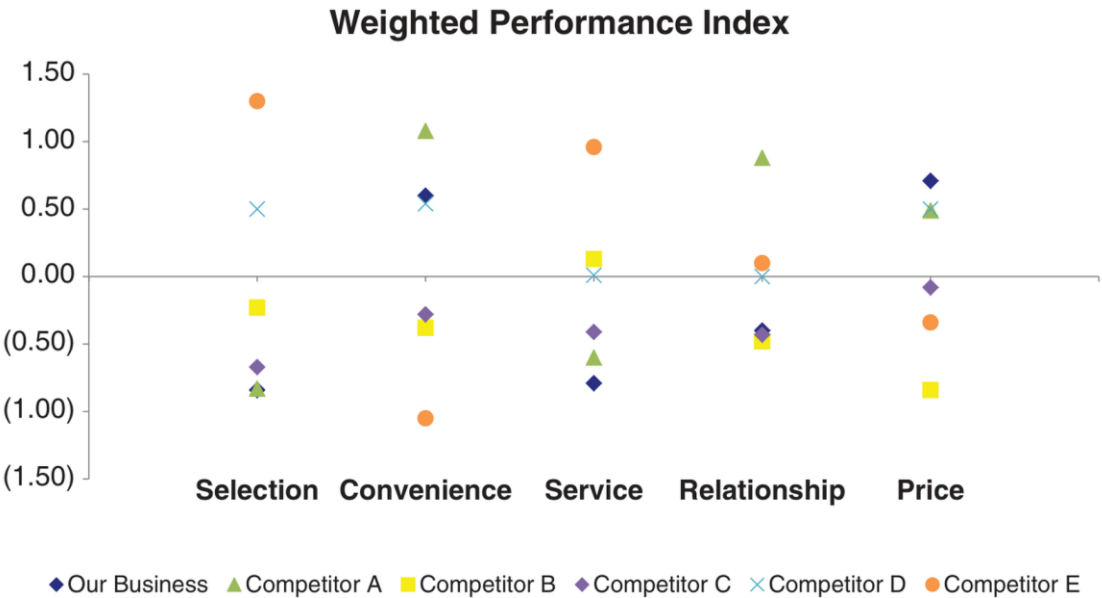
Everything competes for attention.

Buried message

How we compare to competitors in each category is nowhere to be seen.

Takeaway: visual clutter overloads the audience and confuses what they should pay attention to.

Our performance vs. competitors by category



Discuss: how does the right chart improve on the left one by removing clutter?

What is improved and why?



Horizontal bars

One bar per company replaces a scatter of markers. Lengths are easier to compare.

Color contrast

Our business in blue, every competitor in gray.

Positive scale

The numbers are rescaled, because relative differences matter here, not absolute values.

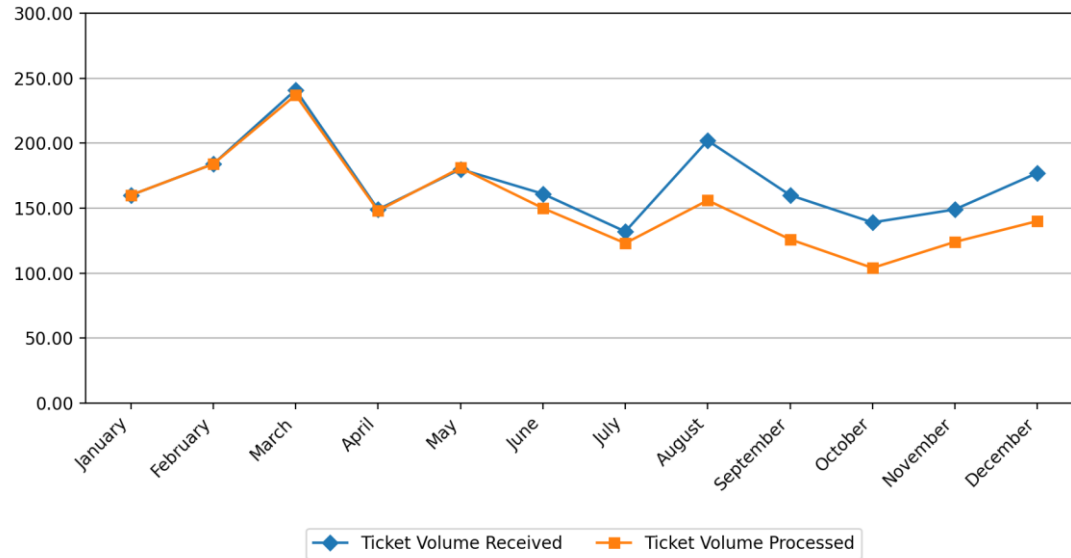
Ordered by our rank

Categories run down from our best (Price) to our worst (Selection).

Rankings labeled

Each bar is labeled with its rank, so we clearly see our position.

Exercise: declutter this chart



Proposition to support: “We lag on ticket processing after May.”

- (a) What clutter does this chart have?
- (b) How would you change it to remove the clutter and show the proposition better?

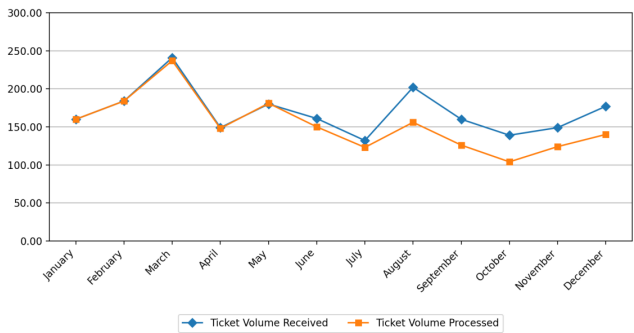
Then use GitHub Copilot in VS Code on the notebook [declutter/exercise-declutter.ipynb](#) to produce the plot you described in (b).

Deliverable: one Google Slide per group with (a), (b) and your improved plot.

Student names: [add your names]

Duplicate this template in Google Slides and add your names: L2 - Remove clutters

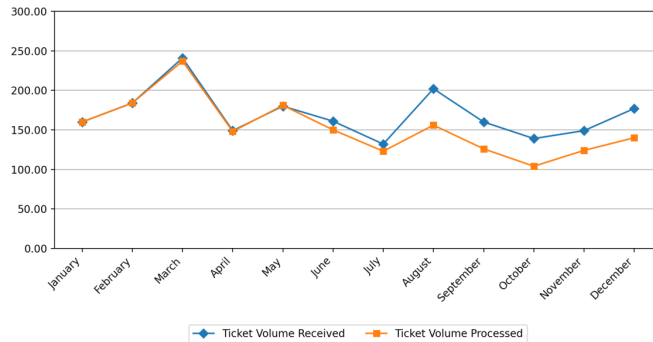
(a) Clutter problems	(b) Suggested changes



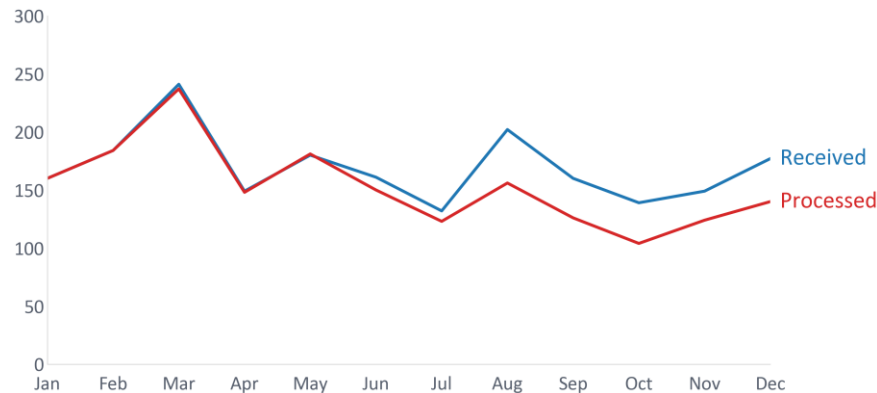
Your improved plot

Better visualization: the same chart, decluttered

Original



Decluttered



1. Remove the chart border

White space frames the visual.

2. Remove the gridlines

The data stands out.

3. Remove the data markers

They add load without information.

4. Simplify the axis labels

No trailing zeros; months abbreviated and horizontal.

5. Label the data directly

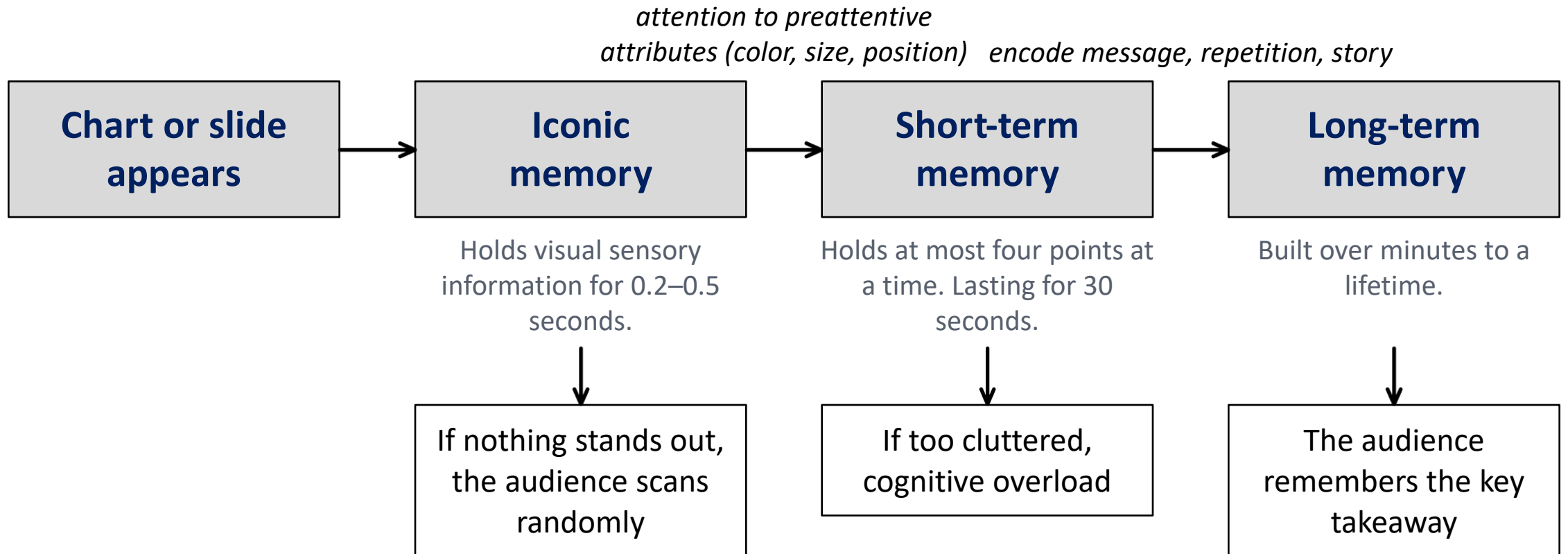
Each label sits next to its line instead of in a legend.

6. Match colors

Each label takes its line's color, so same color means same thing.

Focus attention

How an audience processes a chart



Goal of visual design: guide attention to useful information and encode the key message into memory.

Use color sparingly, and use shades for magnitude

Country Level Sales Rank Top 5 Drugs

Rainbow distribution in color indicates sales rank in given country from #1 (red) to #10 or higher (dark purple)

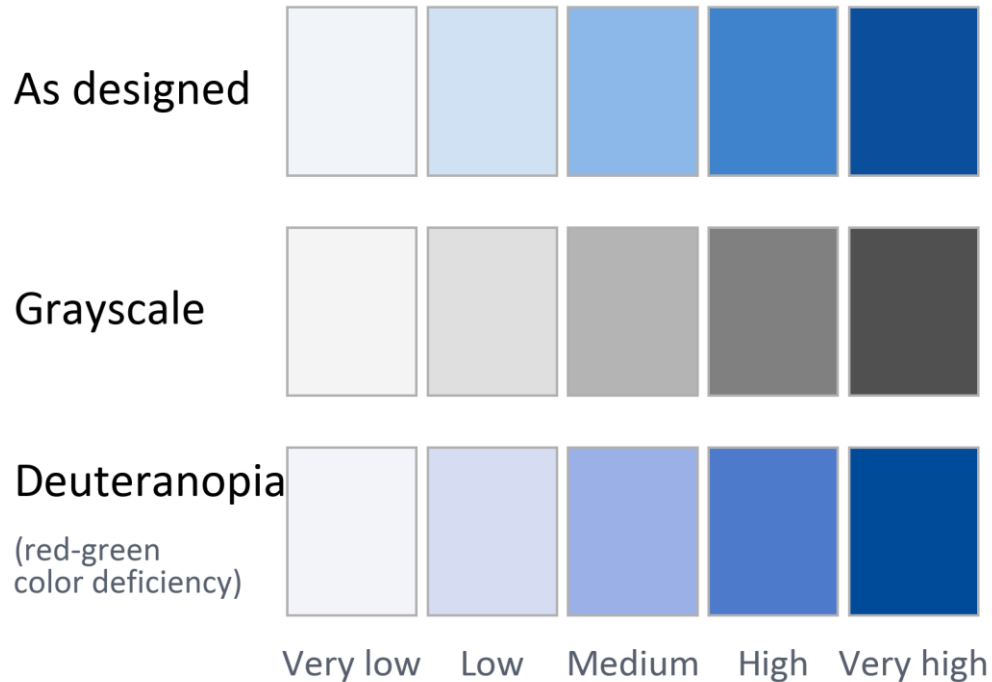
Country	A	B	C	D	E
AUS	1	2	3	6	7
BRA	1	3	4	5	6
CAN	2	3	6	12	8
CHI	1	2	8	4	7
FRA	3	2	4	8	10
GER	3	1	6	5	4
IND	4	1	8	10	5
ITA	2	4	10	9	8
MEX	1	5	4	6	3
RUS	4	3	7	9	12
SPA	2	3	4	5	11
TUR	7	2	3	4	8
UK	1	2	3	6	7
US	1	2	4	3	5

Top 5 drugs: country-level sales rank

RANK		1	2	3	4	5+
COUNTRY DRUG		A	B	C	D	E
Australia		1	2	3	6	7
Brazil		1	3	4	5	6
Canada		2	3	6	12	8
China		1	2	8	4	7
France		3	2	4	8	10
Germany		3	1	6	5	4
India		4	1	8	10	5
Italy		2	4	10	9	8
Mexico		1	5	4	6	3
Russia		4	3	7	9	12
Spain		2	3	4	5	11
Turkey		7	2	3	4	8
United Kingdom		1	2	3	6	7
United States		1	2	4	3	5

- Limit the colors: with too many, nothing stands out.
- Use **shades of one hue** to show magnitude.
- Lightness changes in order: the darker the blue, the better the rank.
- Best when values run from low to high with no meaningful midpoint.

A same-hue palette still reads in grayscale and to color-deficient eyes

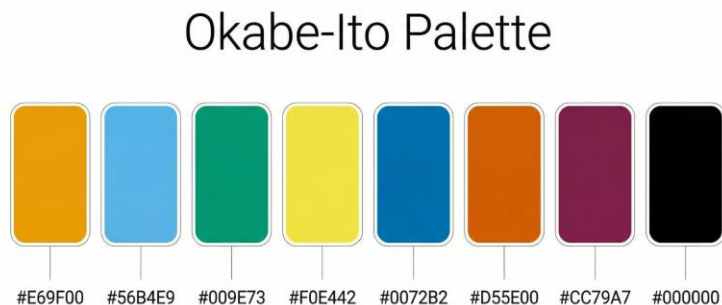


- The order low to high survives in every row, because the palette varies lightness, not hue.
- A rainbow palette's colors would be hard to tell apart in grayscale.

Check: print your chart in grayscale, or run it through a color-deficiency simulator, before you send it.

Accessibility design for color-deficient readers

- The three most common types: deuteranopia and protanopia (red–green; about 8% of men and 0.5% of women) and tritanopia (blue–yellow, rare).
- **Never let red vs. green be the only cue:** add labels, lightness or shape to tell them apart.
- A palette safe for color-deficient readers to start from: Okabe–Ito (siegal.bio.nyu.edu/color-palette).



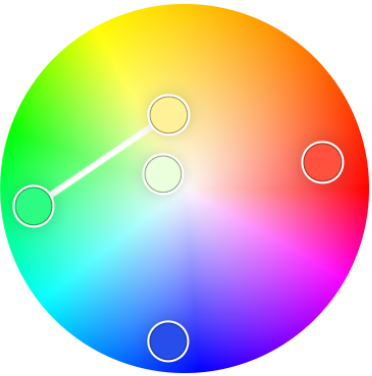
Check a palette with Adobe's color accessibility tool

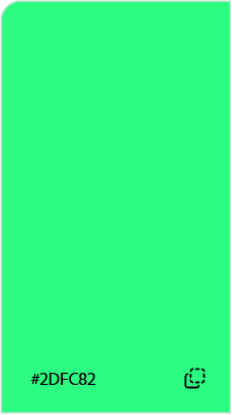
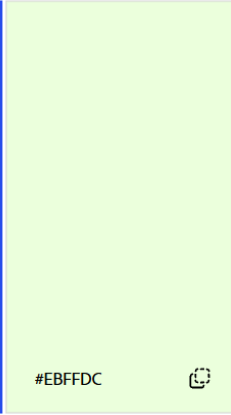
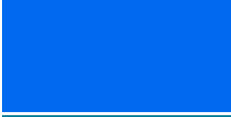
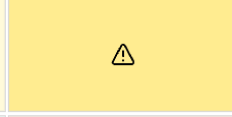
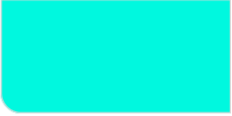
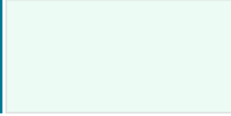
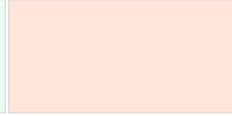
Adobe Express

Color blindness simulator and safe palette tool.

Simulate common types of color blindness and test your color palettes for accessible, inclusive design. Identify potential visibility issues and create color-blind-safe combinations.

Potential color blind conflicts ⚠ Conflicts found



	 #2DFC82	 #274DEA	 #EBFFDC	 #FFF197	 #FF513D
Deuteranopia					
Protanopia					
Tritanopia					

Palette name
My Color Theme

Create with my color palette

Try it: color.adobe.com/create/color-accessibility

Size and enclosure point the eye to what matters



91%

of customers liked the product

Size

The number is far larger than anything else, so the eye lands on it first.

Enclosure

A box sets it apart from the rest of the slide and marks it as the key message.

Summary

1. Decide the goal first: whom you present to, what they should know or do and how the data makes the case. The goal decides which chart is better.
2. Let the question pick the data and the chart: the right metric and grain; an honest axis (a zoomed axis needs the change stated and the longer history); absolute stacks for totals, proportion stacks for shares; avoid pies and 3D, because eyes compare length better than area.
3. Declutter: every element that does not carry the message adds cognitive load and hides the point.
4. Direct attention: color used sparingly, shades of one hue for magnitude, palettes safe for color-deficient readers, size and enclosure.